Directory Services Report/Configuration

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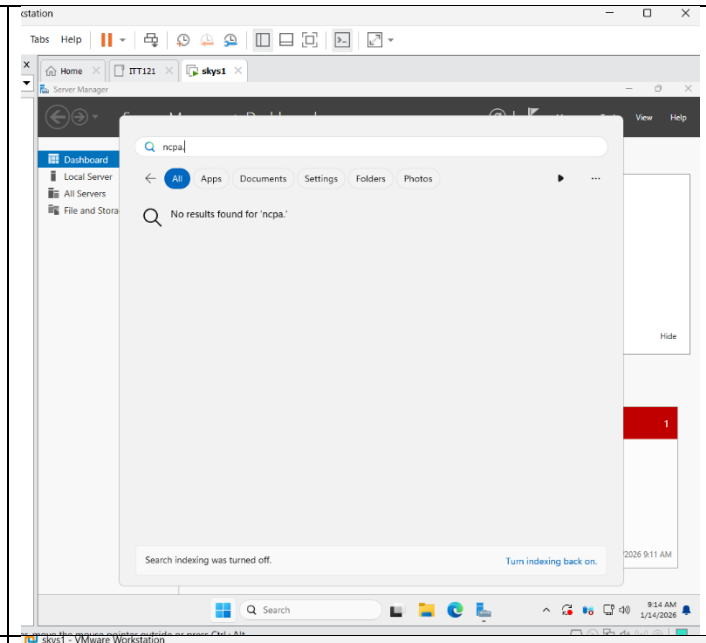
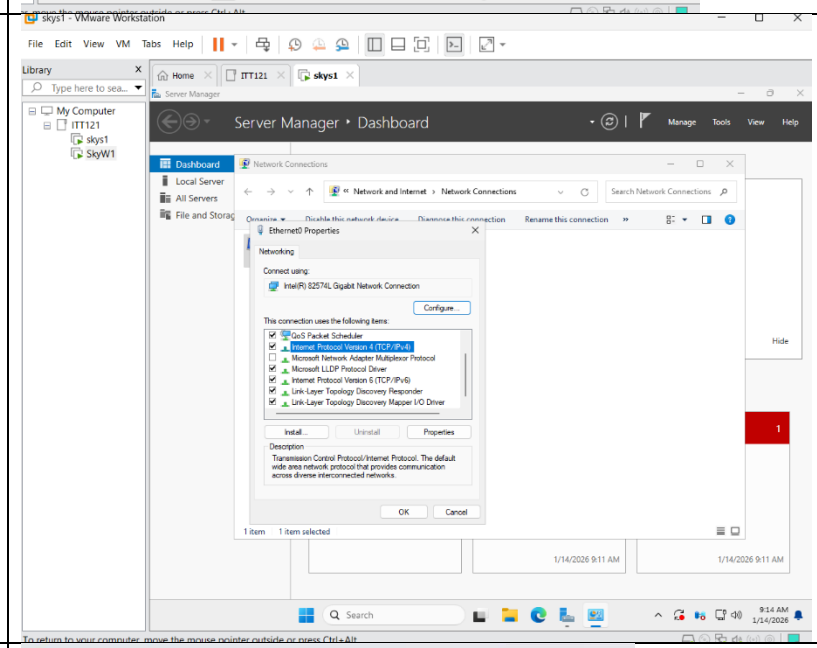
To start our tasks, we are going to configure our IP addresses for the workstation and our server.

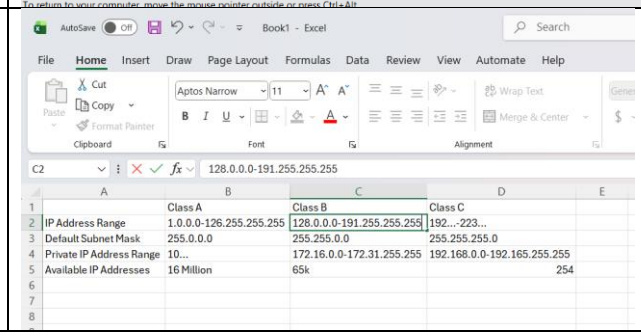
Click the windows icon in your taskbar and type out “ncpa.cpl” and hit enter



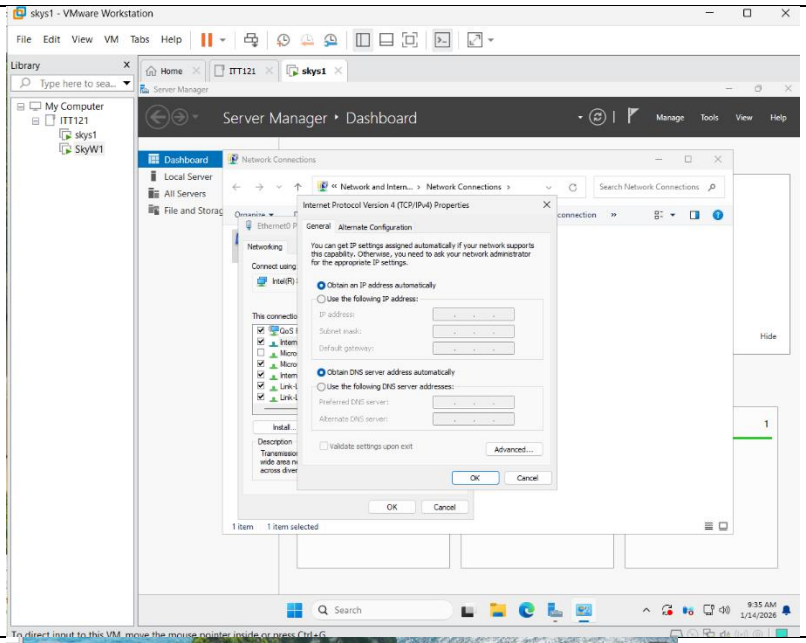
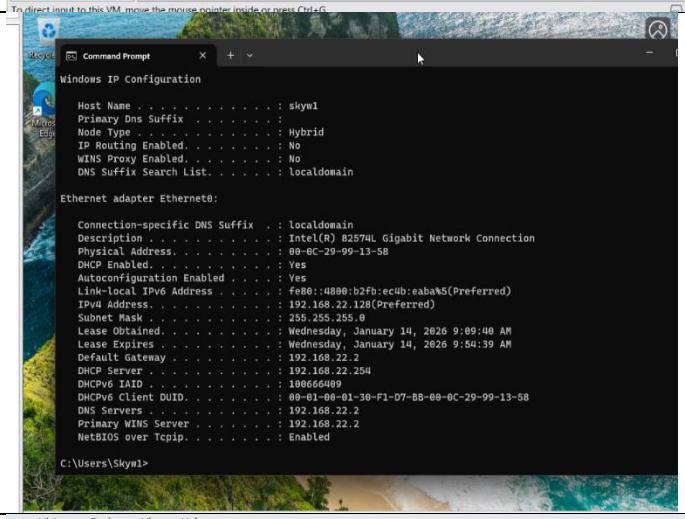
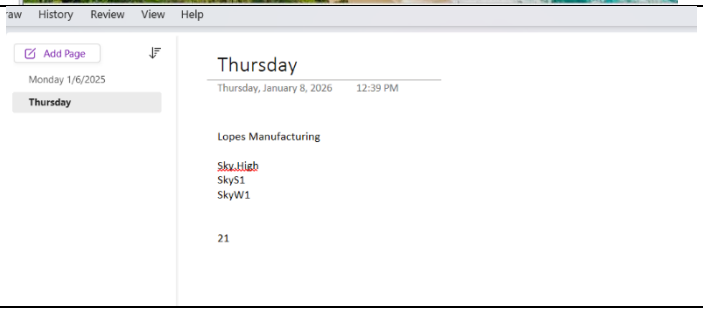
You will be brought to the network connections menu and their should be one option, Ethernet. Right click on ethernet and open up the properties.

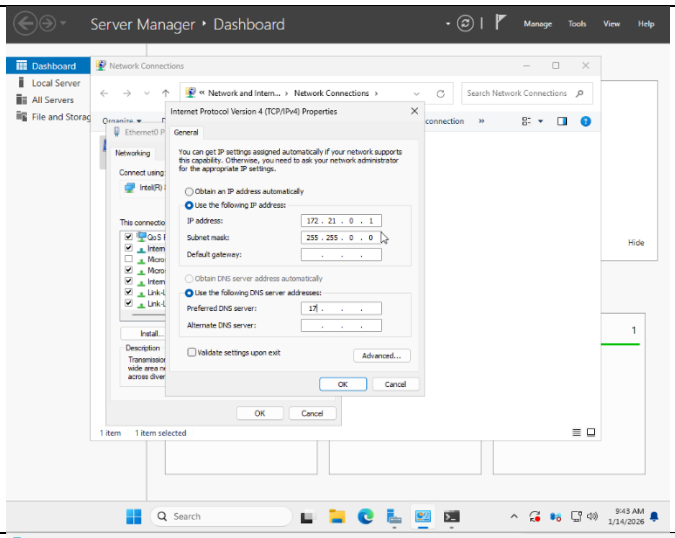
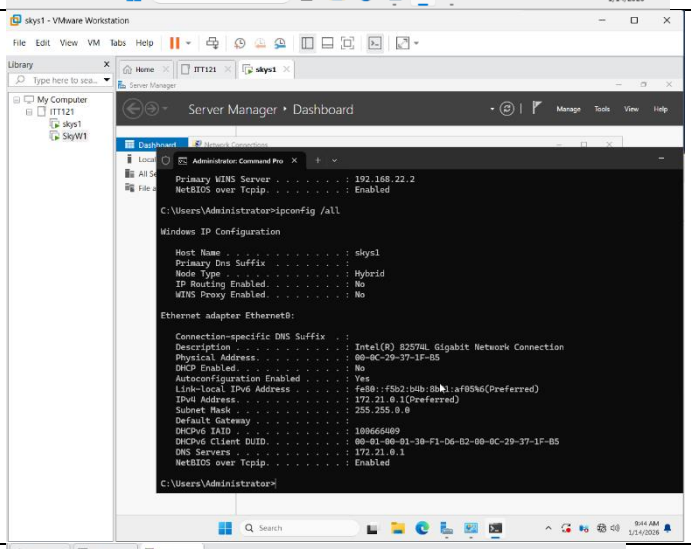
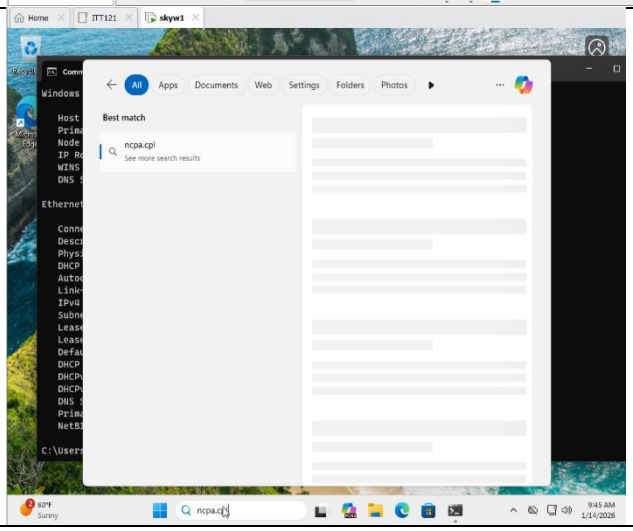


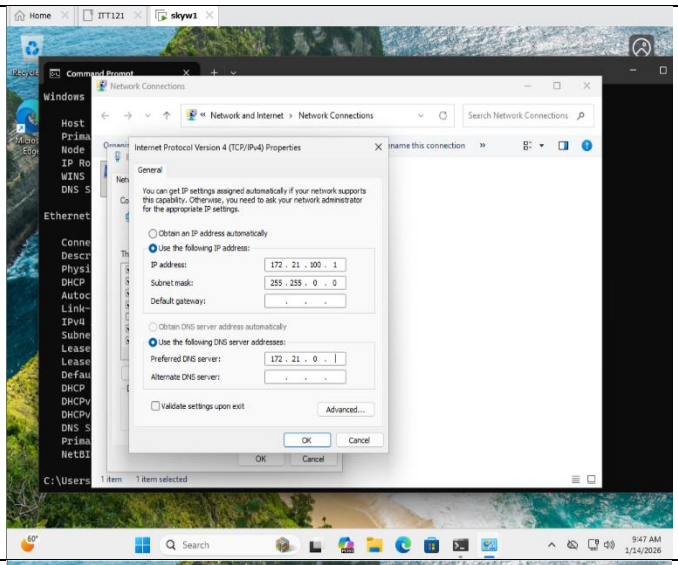
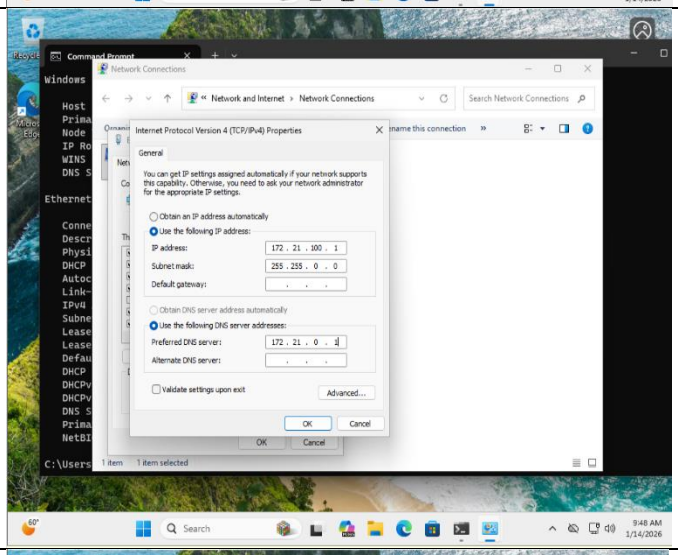
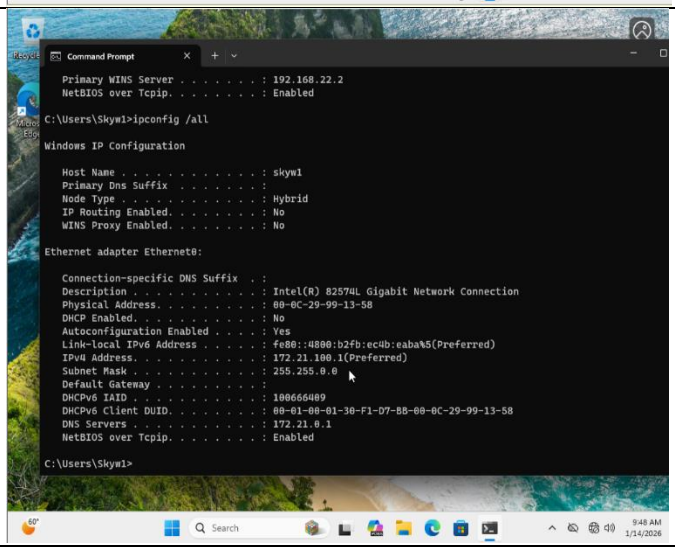
Now before we move on to configure our IP addresses we should familiarize ourselves with IP address ranges, default subnet masks, private IP addresses, and available addresses. This table should help as a basic guide.

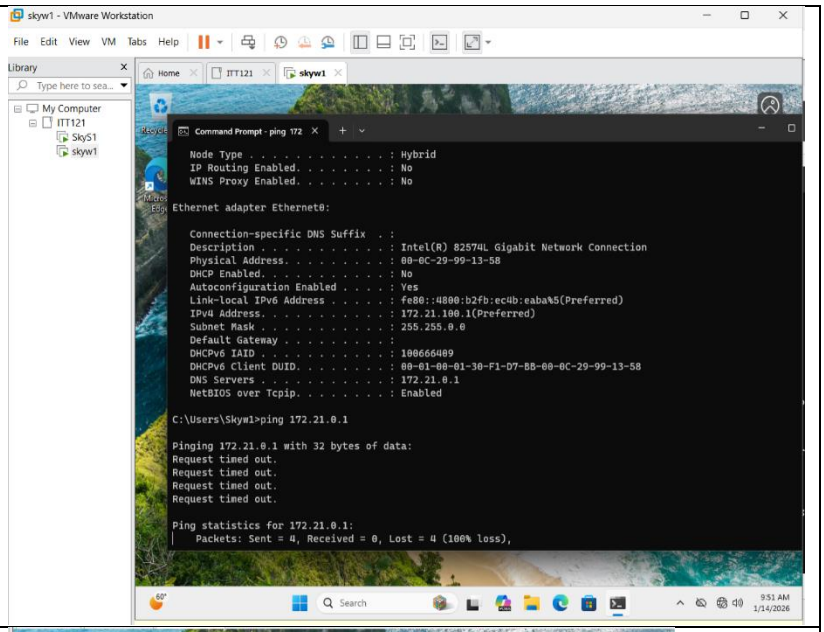
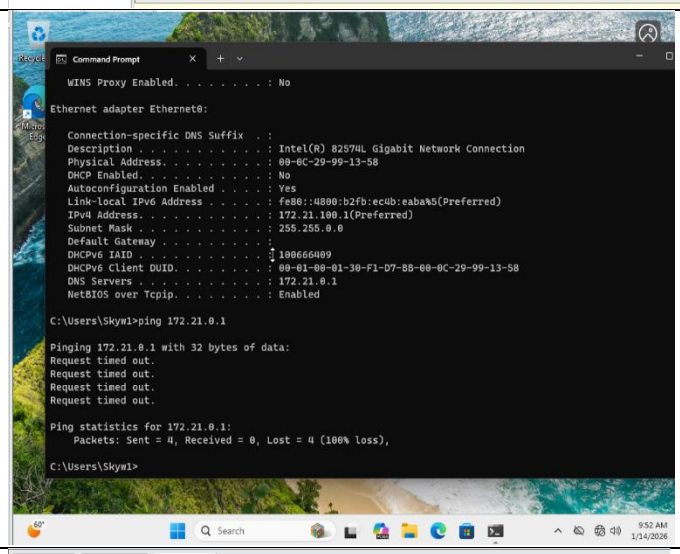
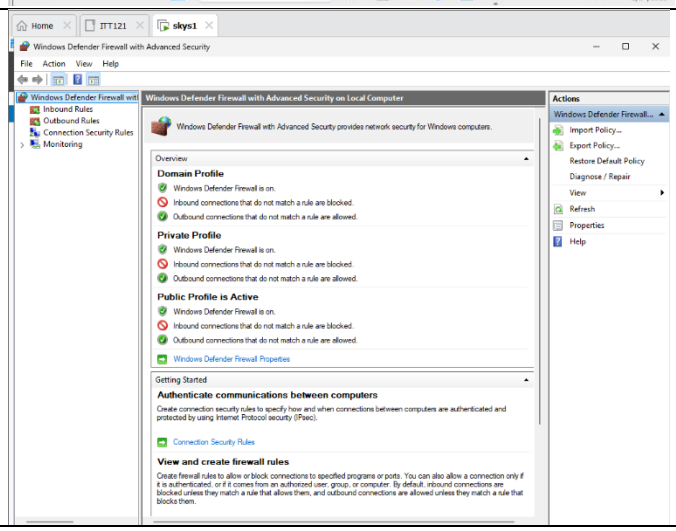


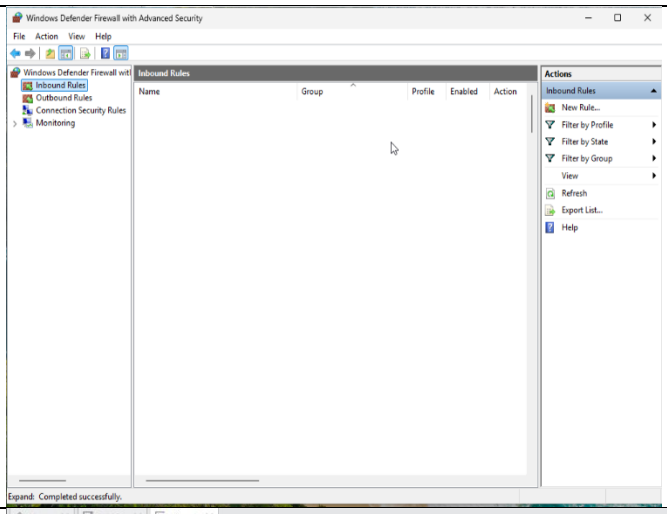
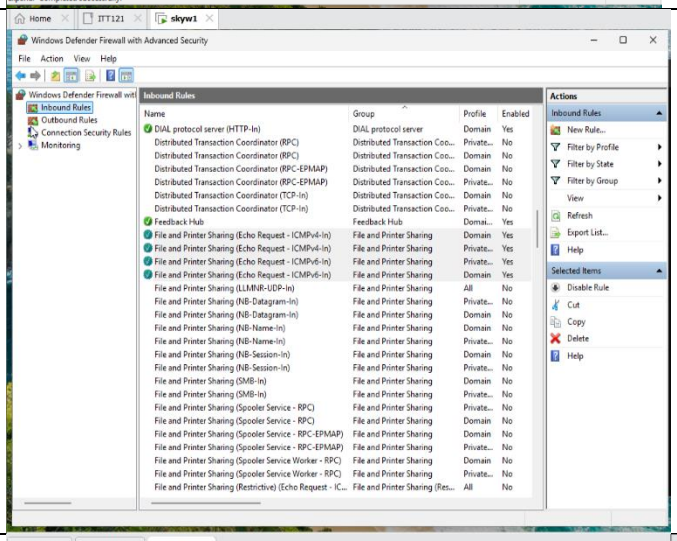
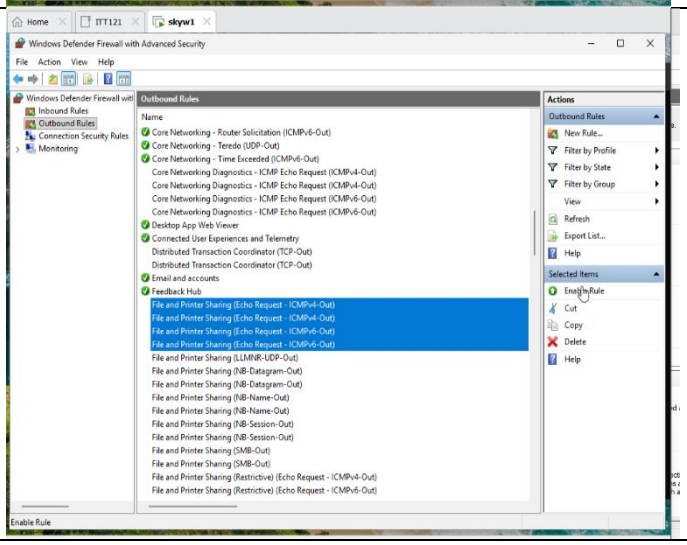
	Class A	Class B	Class C
IP Address Range	1.0.0.0-126.255.255.255	128.0.0.0-191.255.255.255	192...-223...
Default Subnet Mask	255.0.0.0	255.255.0.0	255.255.255.0
Private IP Address Range	10...	172.16.0.0-172.31.255.255	192.168.0.0-192.165.255.255
Available IP Addresses	16 Million	65k	254

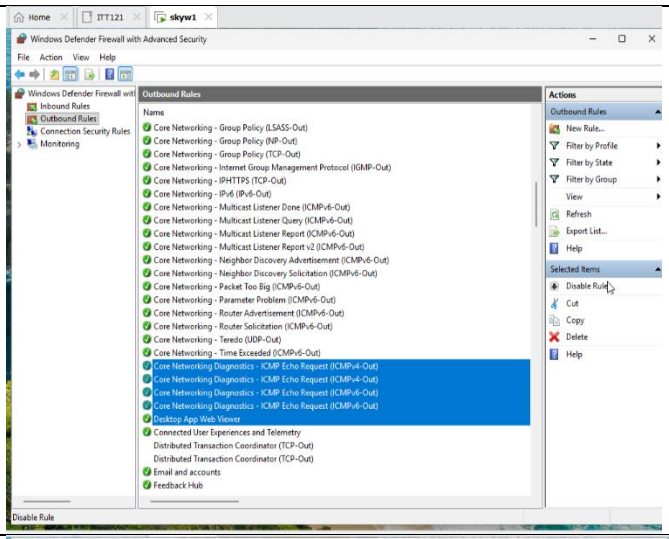
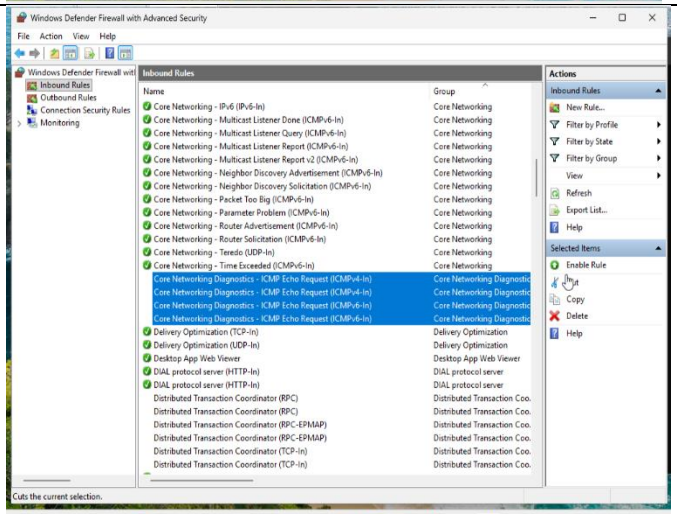
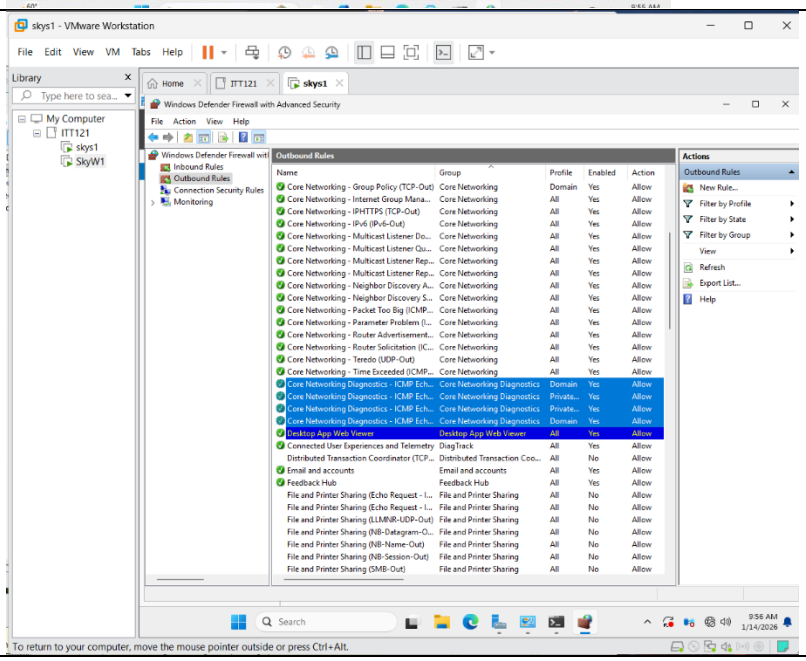
Now going back to our server, find “Internet Protocol Version 4” and click on it to open this menu.	
If you want you can see what your current IP information by typing “cmd” in the windows search bar and hitting enter. Then once in command prompt, type “ip config” or “ip config /all” and hit enter to see IP information or detailed IP information.	
Before we move on, check your notes and find what you assigned number was. We will need this number for IP configuration.	

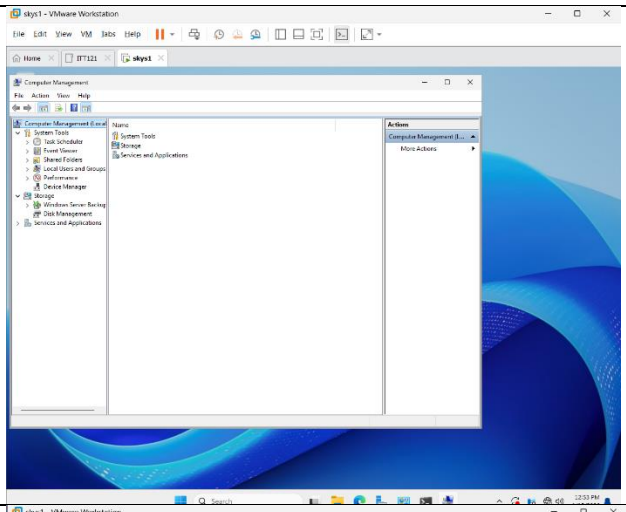
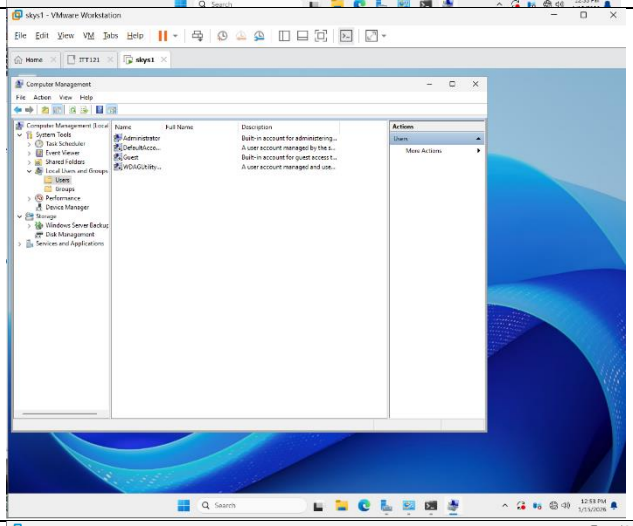
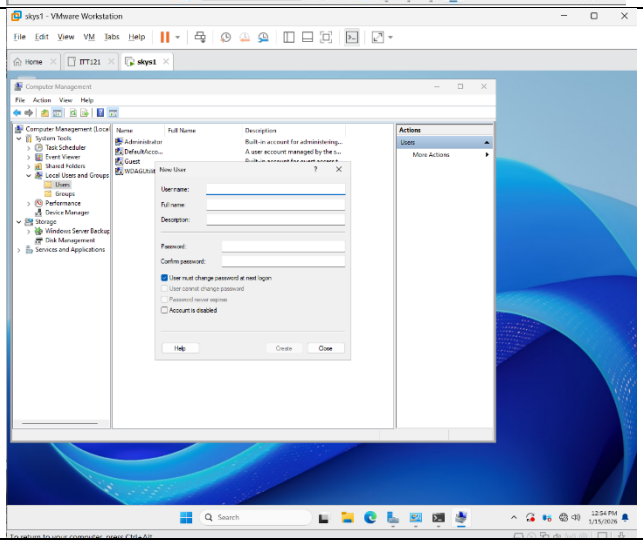
Now return to the server and the IP menu. Select the radio button “use the following address”. For the server IP address it will be: 172.#.0.1 (where # is your number from your notes). Set your subnet to 255.255.0.0 and in the next container, set your preferred DNS server to the same address as the one you just set for IP address.	
After you are done hit “OK” and return to command prompt. Enter the command “ip config /all” and you should now see your new preferred IP address and the same IP address for your DNS server.	
Now switch over to you workstation. The steps will be essentially the same. In the search bar enter ncpa.cpl to view your internet connections.	

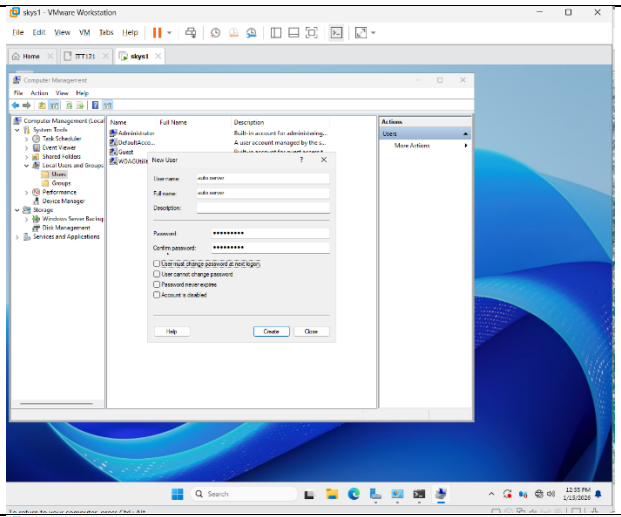
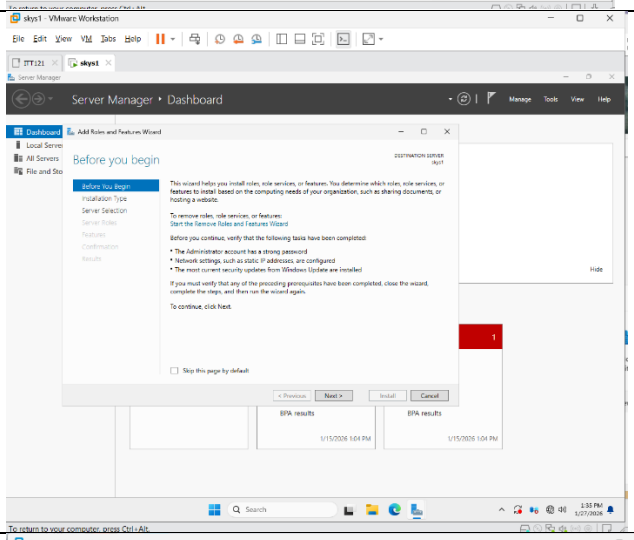
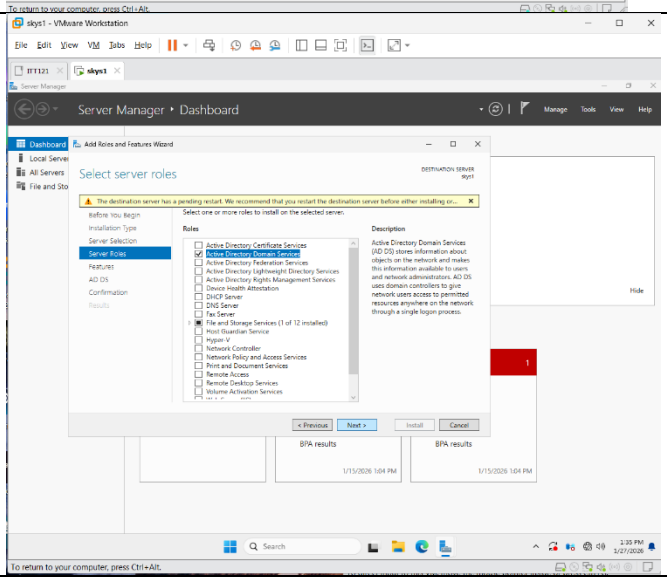
Select Ethernet properties and then “Use the following IP address”. This time the address will be: 172.*.100.1 (Where # is your assigned number). Subnet is 255.255.0.0 For the DNS server option, it should be the same address for the server’s DNS option, 172.*.0.1	
Again make sure that the DNS server is not 172.*.100.1 The DNS needs to be the same address as the one we set for the server. Once you have finished select “OK”.	
You can open command prompt by typing in “cmd” in the windows search bar and pressing enter. Once you are in command prompt we can use the same commands from the server. “ip config” or “ip config /all” to see IP information or detailed IP information. You should see your new preferred address is updated.	

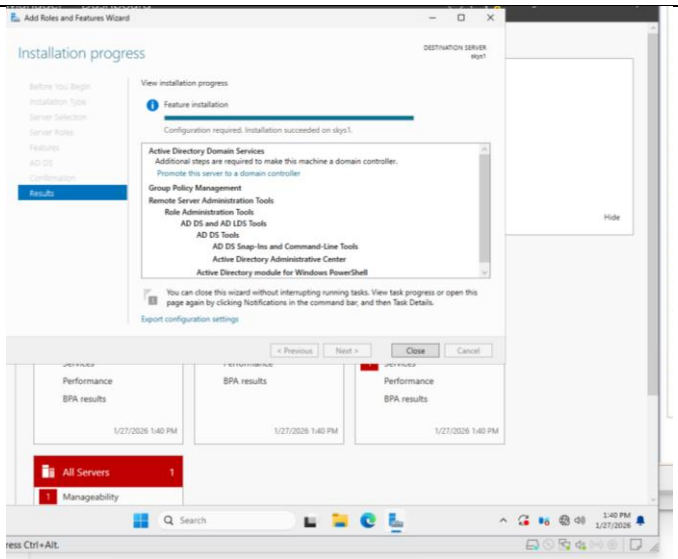
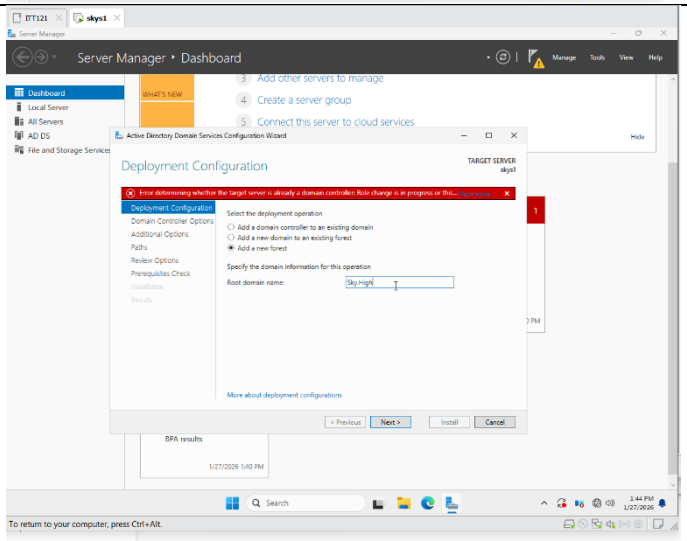
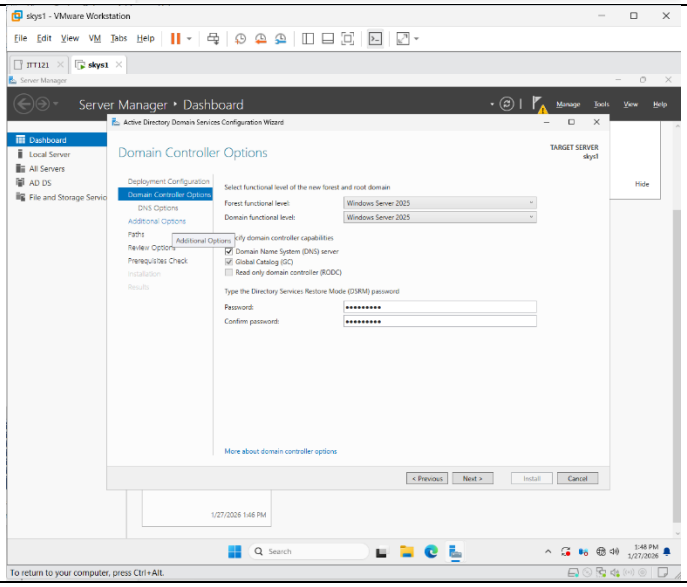
While still in command prompt, go ahead and type in “ping 172.*.0.1” and hit enter. Remember that * is your assigned number. This command attempts to send packets to the server, but you will lose those packets.	
If we do this same command in the server’s command prompt but put in the windows address, 172.*.100.1, we will also lose the packets sent.	
Switch to your server and in the windows search bar look up “Firewall with advanced security” and hit enter.	

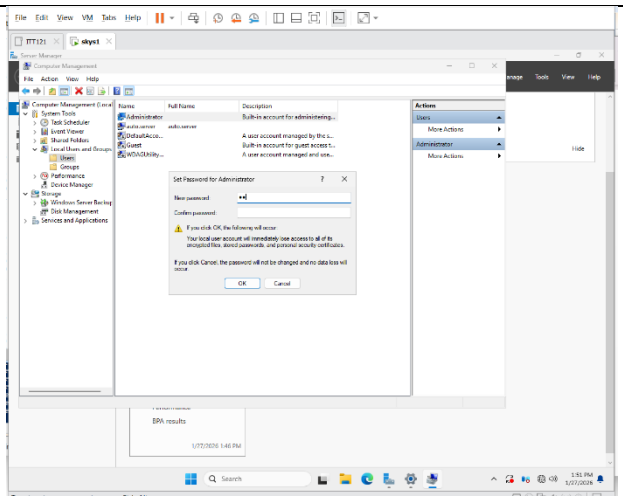
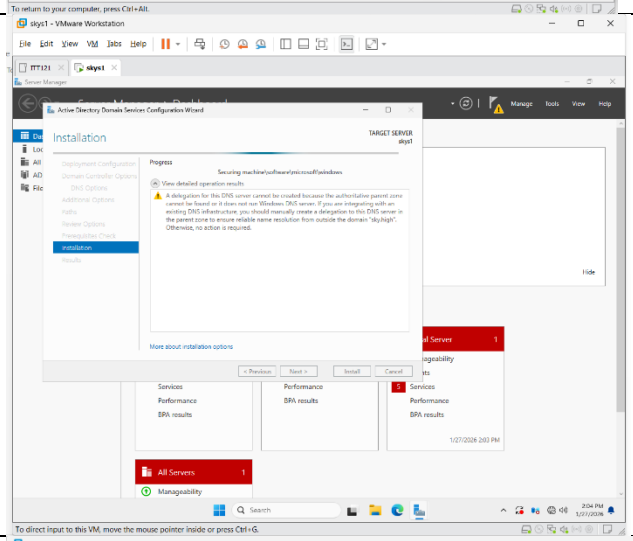
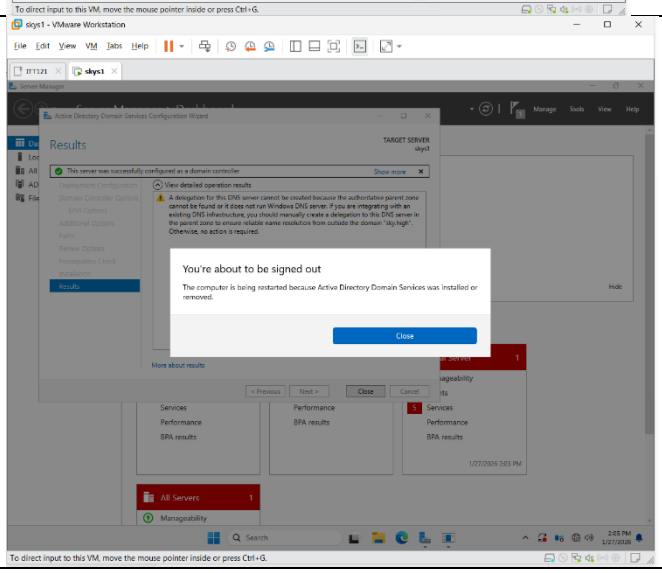
Once you are inside the window we will need to check our rules.	
Select inbound roles and look for “Echo Request ICMPv4”. You can select these and set enabled to “yes”.	
Now in the same windows switch to the out bound rules and look for the same rules with “Echo Request ICMPv4”. Change the enabled state to “yes”.	

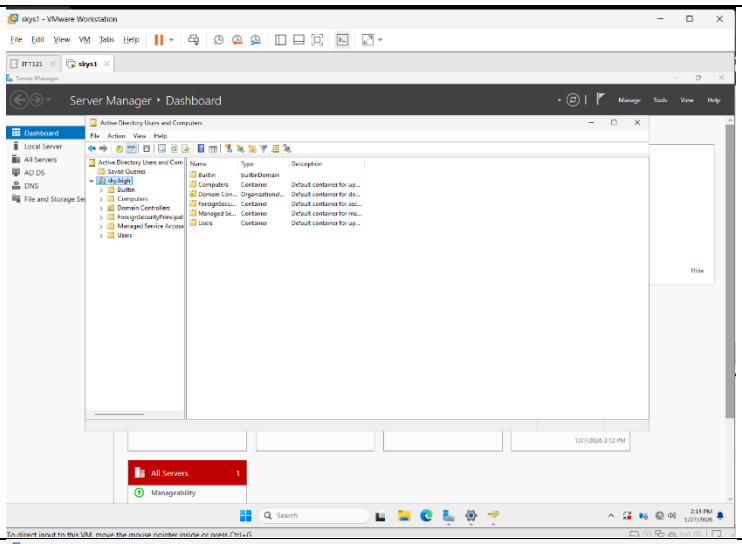
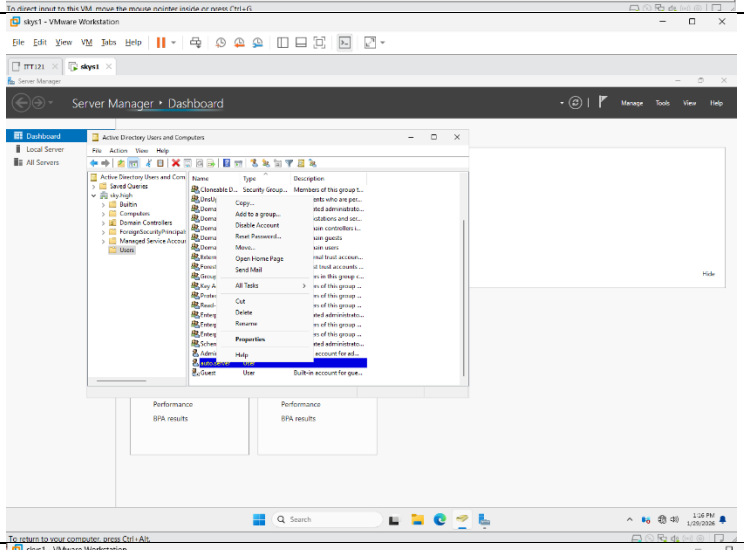
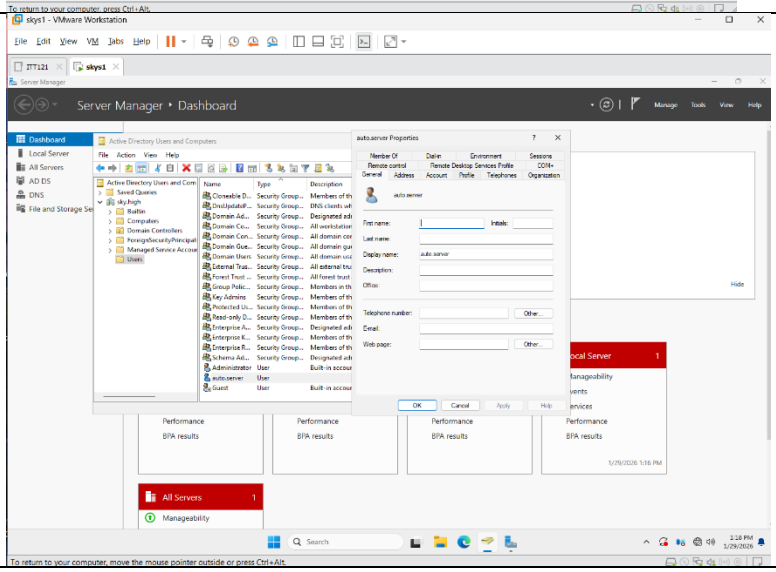
Now we will switch to the workstation and repeat this process. In the windows search bar look up “Firewall with Advanced Security”. Find the Echo request rules and change the enable state to “yes”.	
Make sure to find the inbound and out bound echo requests and change their enabled states to “yes”.	
As a side note, once the rules are enabled you will know they are enabled because when they are enabled, a green check to the left will appear.	

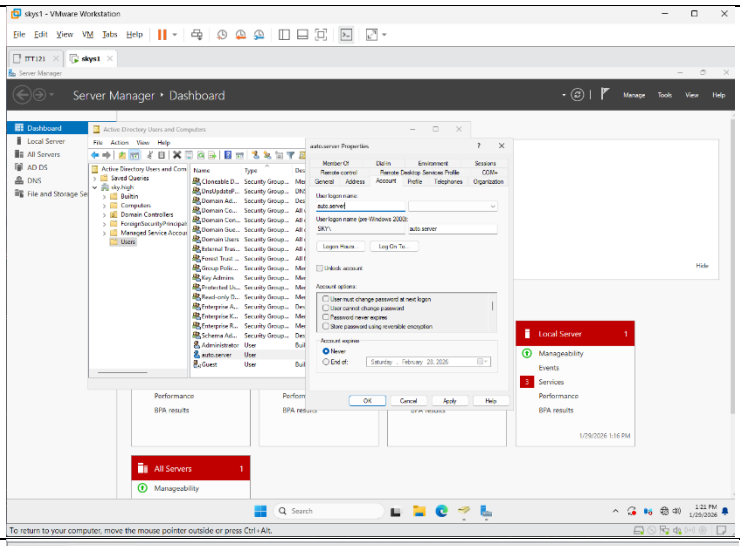
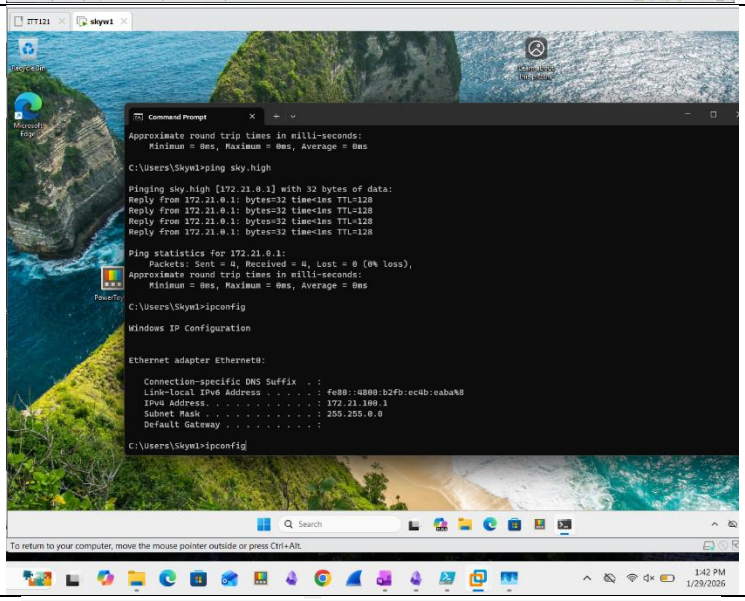
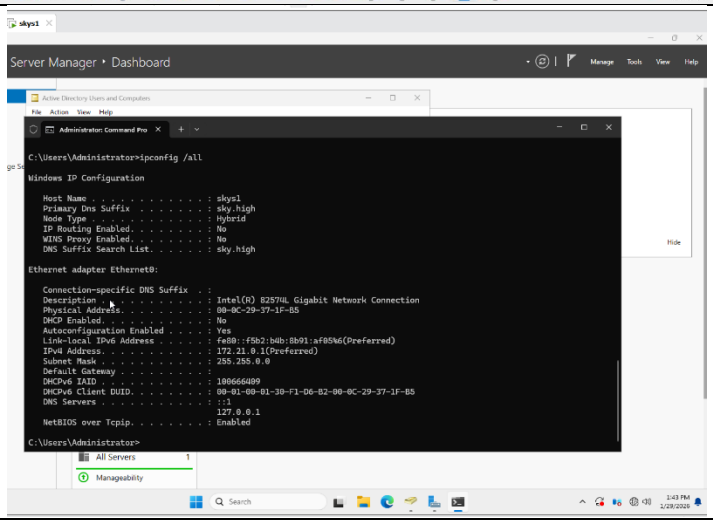
Now we will explore the computer management window and adding users. In the search bar type “comptmgmt” and hit enter. Once you are in the computer management windows, take a moment to look around.	
When you are ready find the local users and groups section and expand it to find the users folder.	
Here in this folder you can view all of the users and by right clicking the white empty space, you can select the option to create a new user.	

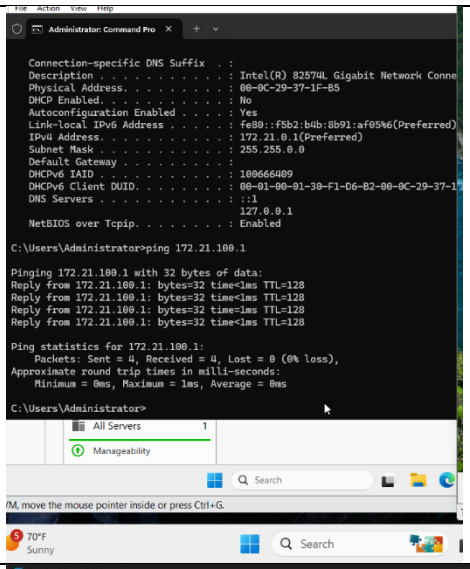
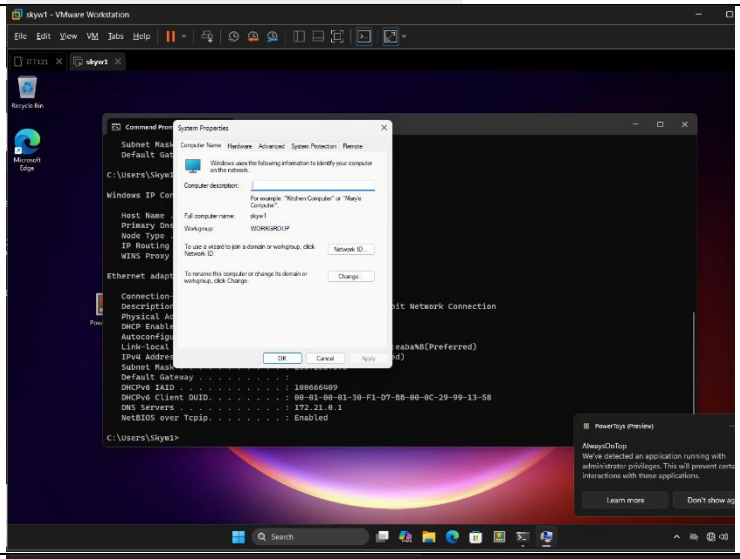
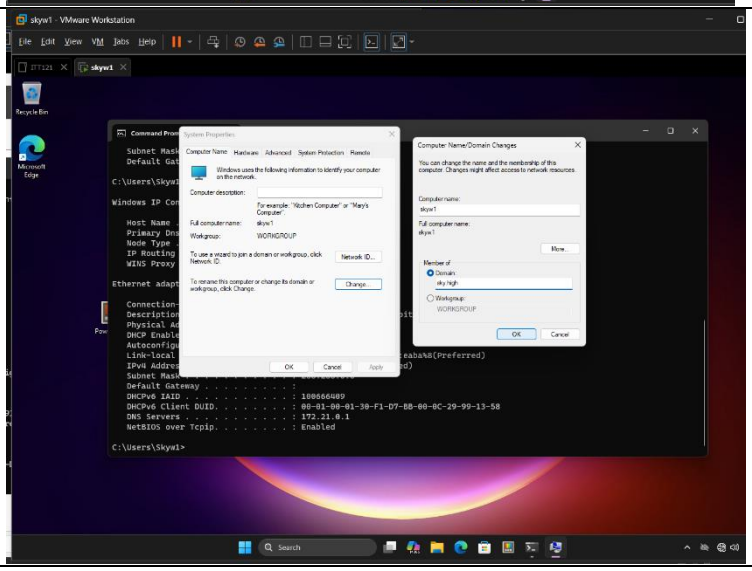
Make a new user and call it “otto.server” (ignore my spelling auto.server). Set the password as Red.vine1 And hit “Create”.	
Once you have created the user we can move onto installing Active Directory. In the server menu you can select “add roles and features” and it will open this page. Before you begin information.	
Click next through the pages until you are the “server roles” page. Select “Active Directory Domain Services” and hit “next”.	

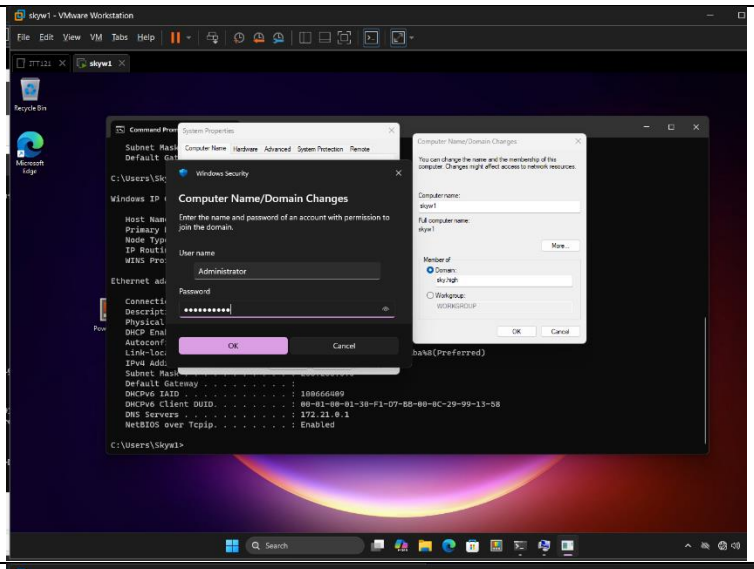
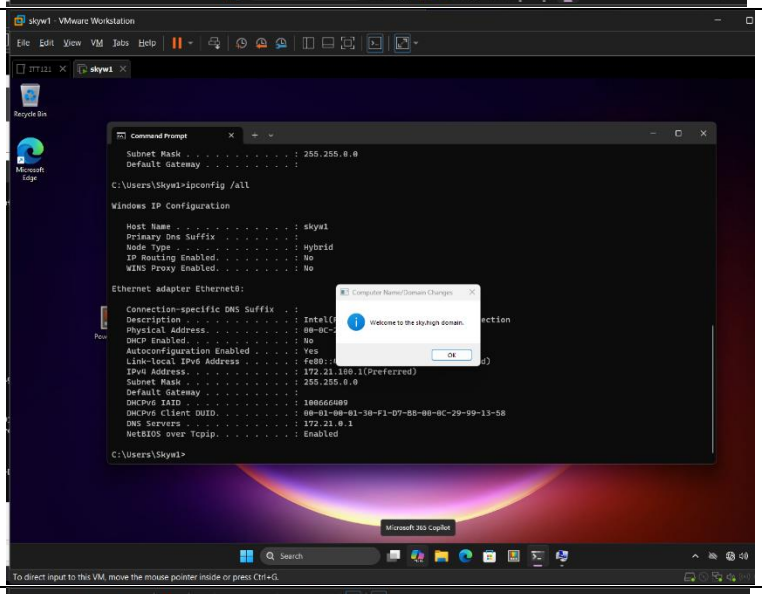
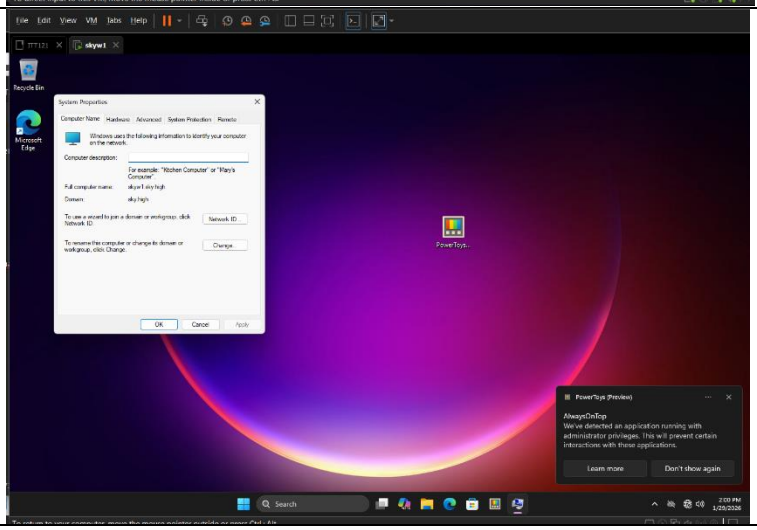
You can continue through the next pages until it is time to install. Go ahead with the installation, once it is done you will need to manually click the “close” button.	
You will be back at the server manager dashboard, and in the top right corner you will see a yield sign underneath the flag. Click on the flag and chooses to configure the AD configuration. You will need to select a domain name, which should also be in you notes. For example my domain name was “sky.high”.	
Once you have the domain name set up, continue through the pages until you are at the DSRM password screen. Set your DSRM password “Red.vine1”.	

You may continue through the rest of the pages, if you get to the end and get a error because there is no administrator password set; go to the windows taskbar and “compmgmt”. Find the users folder again and find the Administrator account. Right click and choose “set password”. The password should be “Red.vine1”.	
Now you should have no errors and should be able to finish the AD services configuration.	
Once you have finished you will be signed out and you will need to restart your server vm.	

Once your server finishes restarting, in the windows search bar search for “Active Directory Users and Computers”. Once you find the correct application you should be able to see your sky high directory with its embedded folders.	
Under the sky.high folder, find user and find your “otto.sever” user. Right click and open otto.server properties.	
You can take some time In the properties to explore all of the properties available for users. When you are ready, find the “account” options.	

In the account options set the user login name as “otto.server”.	
Now we will add the workstation to the domain. Before we do that, go to your workstation and open command prompt. Ping the server and make sure you are getting replies from the server.	
In you server open command prompt and ping the workstation, and make sure that you are getting replies from the workstation.	

In you workstation you should also try pinging the server by DNS. In my case it would be “ping sky.high”.	
In the workstation, in the windows search bar type “sysdm.cpl” and hit enter. In your system properties, find the and select the “change” button.	
In the change menu, select member of domain and type in you domain name. This should be in your notes and in my case it is “sky.high”.	

After typing in your domain name and selecting “OK”, it will ask for an account with permission to join this domain. Use your administrator account and password for this workstation.	
When you have successfully logged in you will have the message box confirming you have joined the domain.	
The workstation will need to restart. Once you are back in the workstation you can type “sysdm.cpl” in the search bar. In your properties window you should now see a domain listed for your workstation. You have now successfully set up Active Directory and joined your workstation to the domain.	

Designing and maintaining effective business information system (BIS) solutions requires strong technological literacy skills across networking, systems administration, cybersecurity, and enterprise applications. Professionals must understand network architecture, IP addressing, and protocols to ensure reliable connectivity between servers, users, and remote sites. Knowledge of operating systems, virtualization, and cloud integration is also essential, as many modern business systems rely on hybrid environments. Additionally, technical literacy includes the ability to assess system requirements, implement security controls, troubleshoot performance issues, and document configurations to support long-term system sustainability.

A core component of many enterprise BIS solutions is Microsoft Active Directory (AD). Active Directory provides centralized authentication, authorization, and management of users, computers, and resources within a Windows-based environment. Key features include domains, organizational units (OUs), group policies, and directory services. Domains establish administrative boundaries, while OUs allow administrators to logically group users and devices for easier management. Group Policy enables consistent enforcement of security settings, software deployment, and configuration standards across the organization. Together, these components support scalability, security, and operational efficiency.

For a corporation with multiple departments and locations, Active Directory is best applied by aligning OUs with business functions such as corporate offices, branch locations, and manufacturing facilities. Role-based security groups can be used to enforce least-privilege access to systems and data, while Group Policy ensures consistent security baselines across all endpoints. Integrating AD with services such as file servers, email platforms, and VPN solutions further enhances centralized control and identity management.

Various information system applications contribute to comprehensive BIS solutions. Enterprise resource planning (ERP) systems support manufacturing, finance, and supply chain operations, while customer relationship management (CRM) systems improve sales and customer service processes. When integrated with directory services and secure network infrastructure, these applications enable efficient data sharing, improved decision-making, and streamlined business operations. Selecting suitable applications requires aligning technical capabilities with organizational goals, scalability needs, and security requirements

References

Active Directory Domain Services overview. (2023) Microsoft. https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview